Test Results — Agentic Finance Reconciliation System

Document version: 1.0 **Test date:** 2026-01-15 **Environment:** n8n · Python 3.11 · Gmail API · AutoCount (staging) · Ubuntu 22.04 **Run command:** `python3 tests/run_reconciliation_tests.py`

> All figures, dates, and account references below are synthetic test fixtures, not client data.

1. Scope

These tests exercise the two pure-logic cores of the system — the **statement parser** and the **matching engine** — plus an end-to-end dry run against a staging mailbox and a staging AutoCount dataset. The goal is to prove that each business rule fires correctly and that no transaction is ever silently dropped.

2. Statement Parser

Fixtures cover three bank layouts (PDF, CSV, Excel) plus deliberately malformed rows.

Case	Input	Expected	Result
PDF table extraction	42-row PDF statement	42 transactions parsed, columns aligned	✓ Pass
CSV column mapping	Bank-B CSV export	Debit/credit split correctly	✓ Pass
Excel with thousands separators	`RM 12,500.00`	Parsed as `12500.00`	✓ Pass
Date normalization	`03/07/2026`, `3 Jul 2026`	Both → `2026-07-03`	✓ Pass
Malformed row	Missing amount column	Row set aside and reported, not dropped	✓ Pass

3. Matching Engine

Each rule is tested with a matching pair and a near-miss that should *not* match.

```
Matching engine — rule verification
Run at: 2026-01-15T04:12:07Z

PASS | Exact match: same date/amount/direction              | reconciled
PASS | Exact match: amount off by 0.01 → not exact           | falls through
PASS | Date-tolerance: amount equal, +2 days (within ±3)     | reconciled
PASS | Date-tolerance: +6 days (outside window)              | exception
PASS | Fee-adjusted: bank charge RM 1.00 within fee range    | reconciled (fee noted)
PASS | FX-adjusted: USD entry matches after rate, within tol | reconciled (FX noted)
PASS | FX-adjusted: rate drift beyond tolerance              | exception
PASS | Transfer pairing: debit/credit pair matched           | reconciled (transfer)
PASS | No candidate → flagged                                | exception
PASS | Two candidates within tolerance → ambiguous           | exception (needs human)
```

Rule	Cases	Pass	Fail
Exact match	2	2	0
Date-tolerance	2	2	0
Fee-adjusted	1	1	0
FX-adjusted	2	2	0
Transfer pairing	1	1	0
No-match / ambiguous	2	2	0

4. End-to-End Dry Run (Staging)

A staging mailbox seeded with 3 client statements was run against a staging AutoCount dataset.

```
Reconciliation dry run — staging
Clients processed: 3
Statements parsed: 3  (128 transactions total)

Client A: 51 txns → 48 reconciled, 3 exceptions (2 unmatched bank, 1 ambiguous)
Client B: 40 txns → 40 reconciled, 0 exceptions
Client C: 37 txns → 34 reconciled, 3 exceptions (1 FX drift, 2 unmatched entry)

Audit log written: 3 run records
Exception alerts dispatched: 2 (WhatsApp + email)  [staging sink]
```

Verified: matched + exception counts always sum to the parsed total (no dropped rows); every reconciled transaction carries the rule that matched it; every exception carries a suspected reason.

5. Summary

Area	Cases	Pass	Fail
Statement parser	5	5	0
Matching engine	10	10	0
End-to-end dry run	3	3	0
Total	**18**	**18**	**0**

All checks passed. The parser handles multiple bank layouts and preserves malformed rows for review, every matching rule fires correctly and rejects near-misses, and the end-to-end run reconciles cleanly while accounting for 100% of parsed transactions with a full audit trail.